Install guide — Field Box BT

App v0.11.0 · box firmware 0.4.1-bt

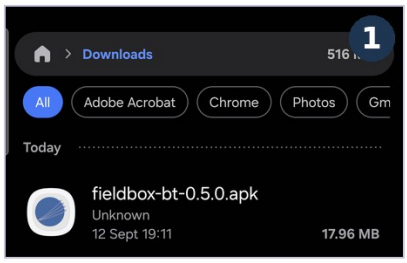
1. What you need

- An **Android 10 or newer** phone, with GPS and Bluetooth.
- The install file **`fieldbox-bt-<version>.apk`** (for example `fieldbox-bt-0.11.0.apk`).
- A **Field Box** with firmware **0.4.1-bt** (boxes on 0.3.1-bt work too, without the extra sensors; older boxes on 0.2.0-bt still work, but without continuous background logging — see §5).

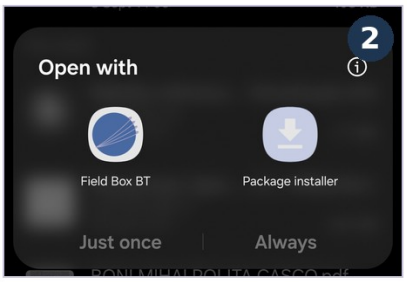
2. Installing the APK on the phone

The app is not on the Play Store; it is installed from an **APK** file sent by the coordinator. The steps below are from a Samsung phone on Android 14–16; on other phones the wording may differ slightly, the order is the same.

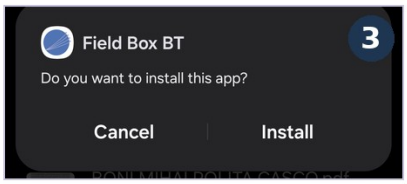
1. Get the file `fieldbox-bt-<version>.apk` onto the phone — via WhatsApp, a USB cable or Google Drive.
2. **Received on WhatsApp:** tap the file right in the chat (the first tap downloads it, the second opens it) — no need to go to Downloads; continue with step 3. **Received another way:** open **My Files → Downloads** and tap the file.
3. If an **"Open with"** window appears, choose **Package installer** (not Field Box BT — that is the app already installed, not the installer).
4. At **"Do you want to install this app?"** tap **Install**. The first time Android may ask to allow installs from that source (WhatsApp, My Files or the browser, if you tapped **Download** in the app): a settings screen with **"Allow from this source"** appears — switch it on, go back and tap the file again.



My Files → Downloads: tap fieldbox-bt-0.5.0.apk



"Open with": choose Package installer

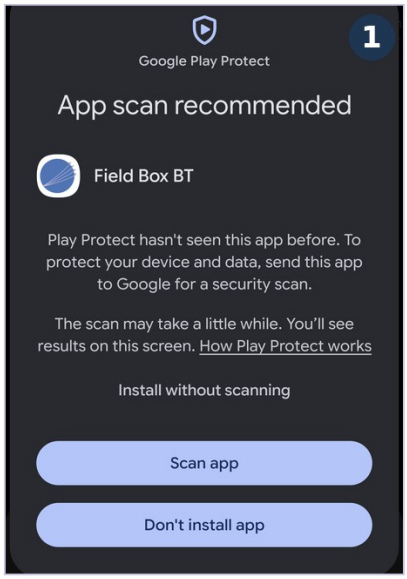


"Do you want to install this app?" → Install

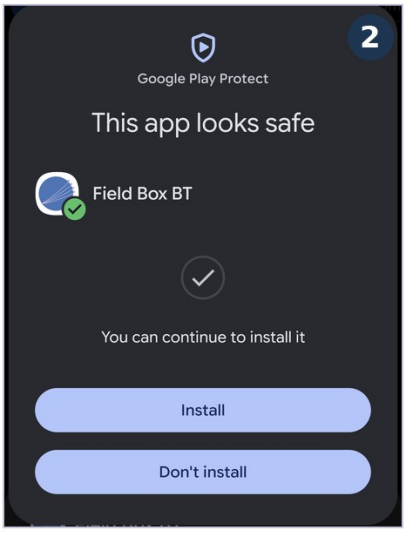
Steps 2–4

5. **Google Play Protect — "App scan recommended"** appears: the app is not from the Play Store, so Google does not know it. Tap **Scan app**; the scan takes a few seconds. Without internet, tap **More details → Install without scanning**.
6. At **"This app looks safe"** tap **Install**.

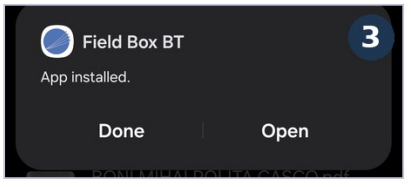
7. At **"App installed"** tap **Open**.



Google Play Protect: tap Scan app (or "Install without scanning" with no internet)



"This app looks safe" → Install



"App installed" → Open

Steps 5–7

The app appears on the phone as **Field Box BT**. A newer version is installed the same way, **over** the old one — the profile, files and bundles on the phone stay. The installed version is shown on the **Boxes** tab, under the connection status (`App v0.11.0 · firmware v…`). When a new version is out and the phone has internet, the same tab shows `FieldBox … is out` with a **Download** button: it downloads the `.apk` itself, which you open from the browser's notification and install as above.

3. First launch

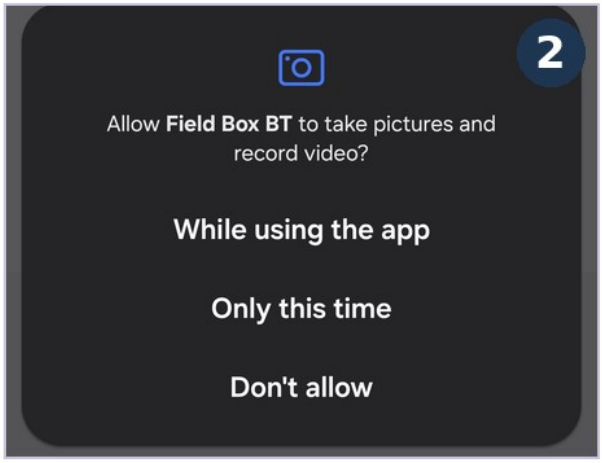
On first open the app asks, in turn:

1. **Profile** — the pupil's/teacher's name and class/school (optional). They appear in the report and in file names. Tap **Save**, or **Skip** (the session is then identified by the box code only). Can be changed any time with **edit** on the Boxes tab.
2. **Camera** — "Allow Field Box BT to take pictures…" → **While using the app** (for the station photos).
3. **Location** — choose **Precise** (not Approximate!) and **While using the app**. Without "Precise" the route cannot be recorded.

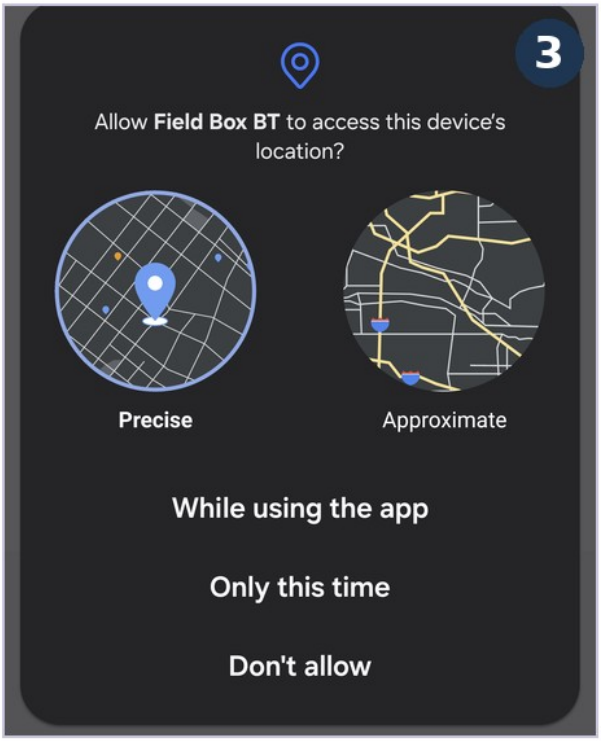
4. **Notifications** → **Allow** (the permanent notification shows the link state and has the **Read now** button).



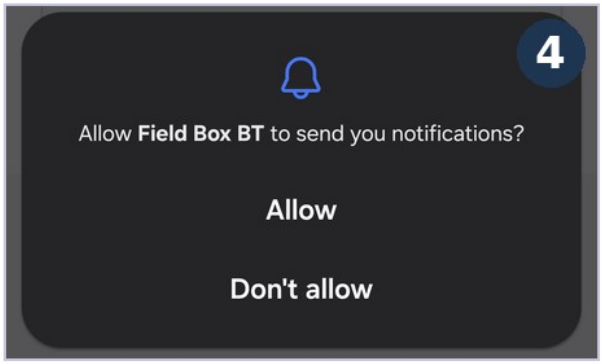
Profile: name and class (or Skip) → Save



Camera → While using the app



Location → Precise + While using the app



Notifications → Allow

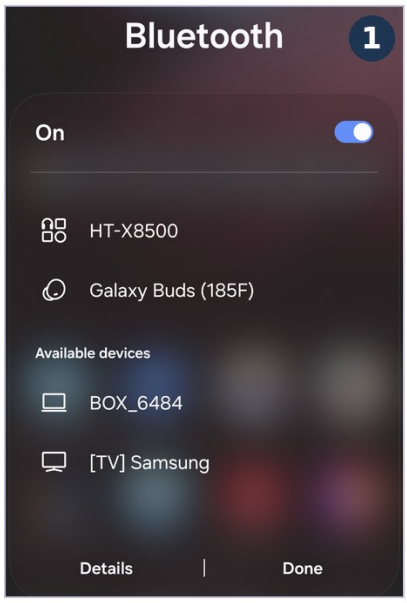
First launch: profile and permissions

Bluetooth pairing (once, per phone and box)

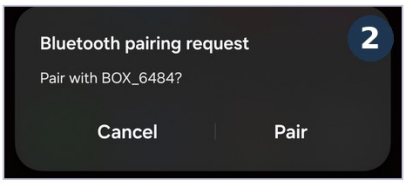
The box must be **paired** with the phone before the app can see it.

1. Power the box (the LED blinks). On the phone: **Settings → Bluetooth** (or long-press the Bluetooth icon in the quick settings).
2. Under **Available devices** the box appears as **BOX_xxxx** (the code on its label) — tap it.
3. At **"Bluetooth pairing request — Pair with BOX_xxxx?"** tap **Pair**. No PIN is asked. The box moves to "Paired devices"; the pairing is kept.
4. In the app, **Boxes → Connect to a box**. The first time Android asks for the **Nearby devices** permission → **Allow**.

5. In the **Choose a box** window tap **BOX_xxxx**. Within a few seconds the strip at the top turns green: **BOX_xxxx · link good**, and the card shows the box's firmware and clock (synchronised automatically).



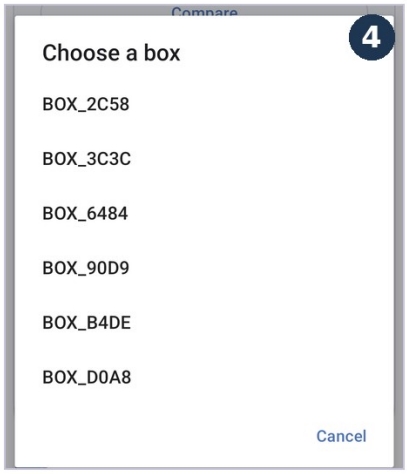
Settings → Bluetooth: tap BOX_xxxx under Available devices



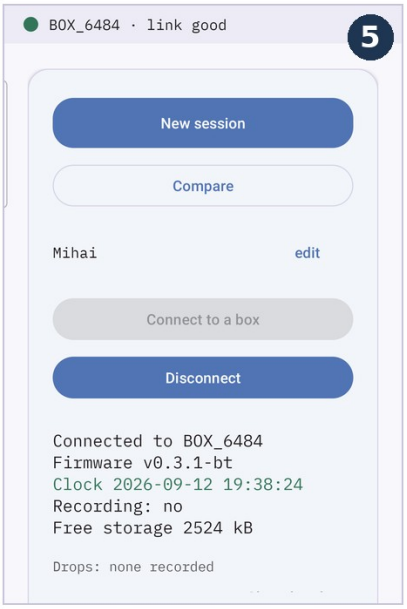
"Bluetooth pairing request" → Pair (once per phone and box)



In the app, Connect to a box → Nearby devices → Allow



Choose a box: tap BOX_xxxx



Connected: green dot, "link good", firmware and clock

Bluetooth pairing and the first connection

4. Quick check

Before heading out to the field, verify:

- The **Live** tab shows sensor values (not blank) once connected.
- The **Point** tab, held outdoors for 1–2 minutes, reaches a GPS fix (the pill turns green).

- The **Map** tab shows map tiles (not just grey), if the area has had signal at some point.

5. Updating the box (firmware)

The box's program (the firmware) is written from a Windows computer, over the USB cable, with the **Arduino IDE**. Every step below has a picture: the red frame shows where to click, and the mouse arrow rests on it.

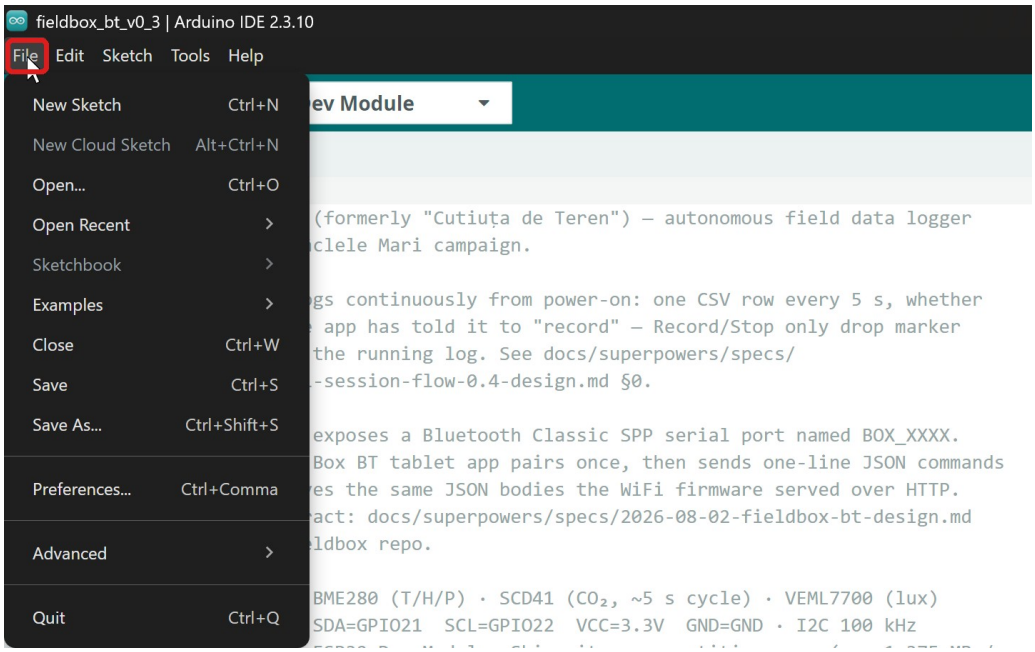
First of all: get the data off the box. The first boot after a firmware flash reformats the box's storage. In the app, **Files** tab: **Download** on each row, or **PDF report** (which fetches the day's log by itself). Anything still on the box is erased.

5.1 What you need

- A **Windows 10/11** computer with internet access (for the one-time setup only).
- **Arduino IDE 2**, free from **arduino.cc → Software**, installed with the default options.
- A **USB data cable** (not a charge-only one) for the board inside the box. If Windows does not see the board, install the **CH340** driver (search for "CH340 driver" on the maker's site, WCH).
- The **`firmware/fieldbox_bt_v0_4`** folder from the FieldBox package (the unzipped ZIP, e.g. in **Downloads**). It holds two files, **`fieldbox_bt_v0_4.ino`** and **`partitions.csv`**; keep them together.

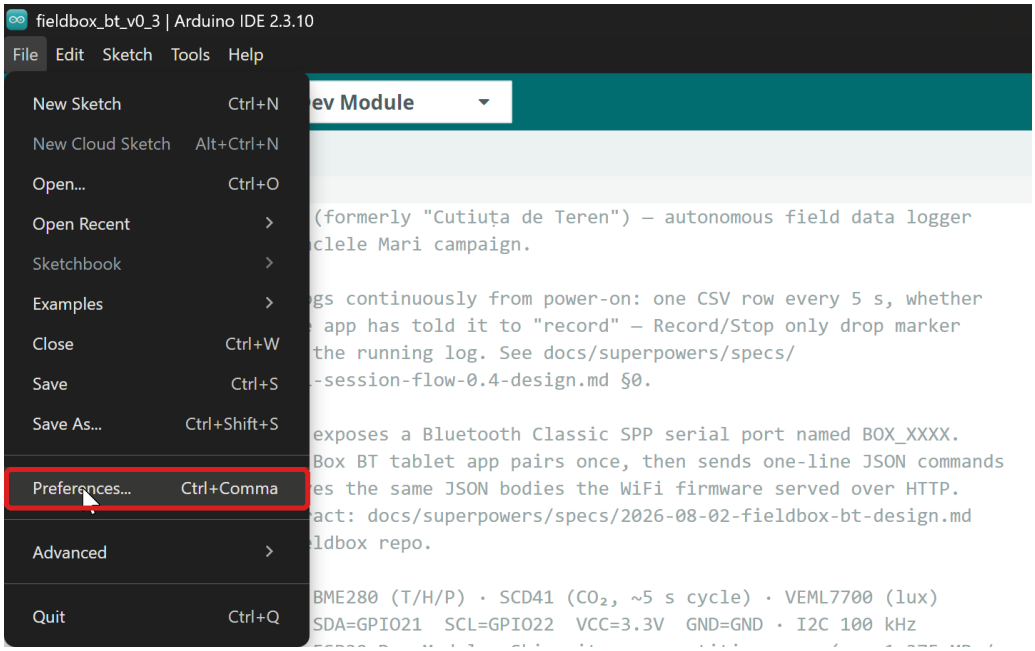
5.2 Setting up the Arduino IDE (once per computer)

1. Open the Arduino IDE and click the **File** menu.



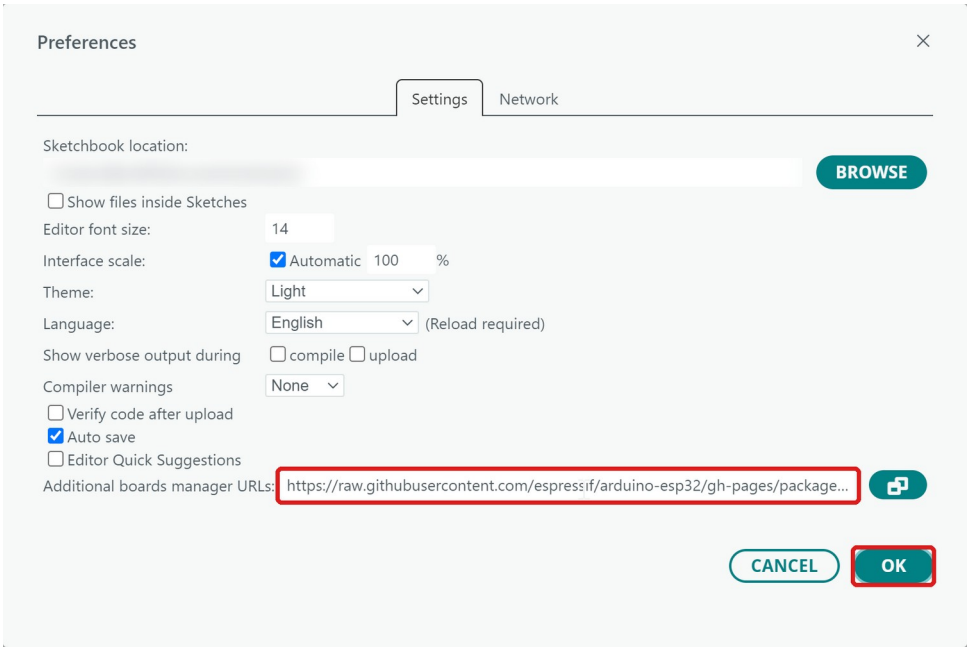
The File menu

2. Choose **Preferences…**.



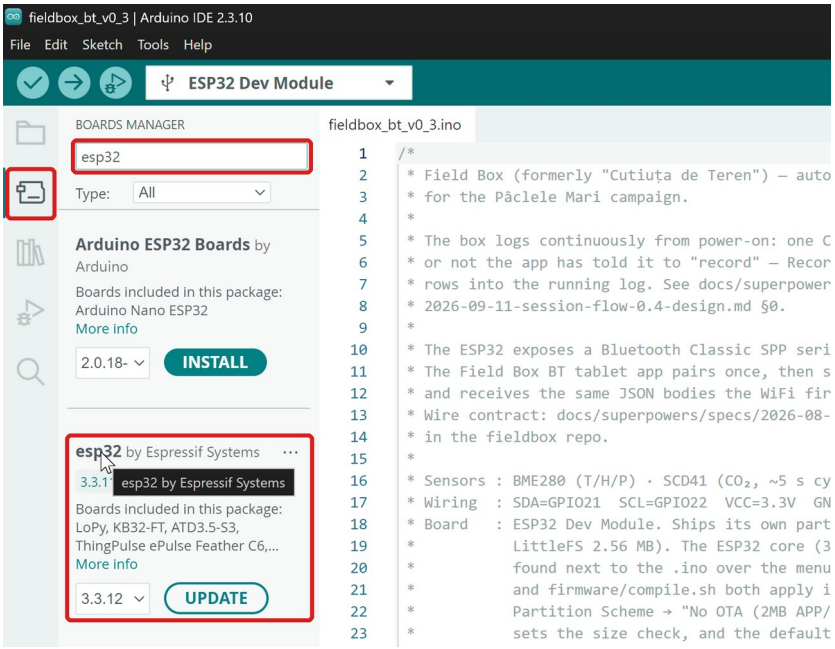
File → Preferences

3. In **Additional boards manager URLs** paste the address below, then click **OK**:
`https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json`



The ESP32 boards address in Preferences

4. Click the second icon in the left bar (**Boards Manager**), type **esp32** in the search box and, on **esp32 by Espressif Systems**, click **INSTALL**. Do not pick "Arduino ESP32 Boards" above it. It takes a few minutes. If it says "installed" under the name, there is nothing to do.

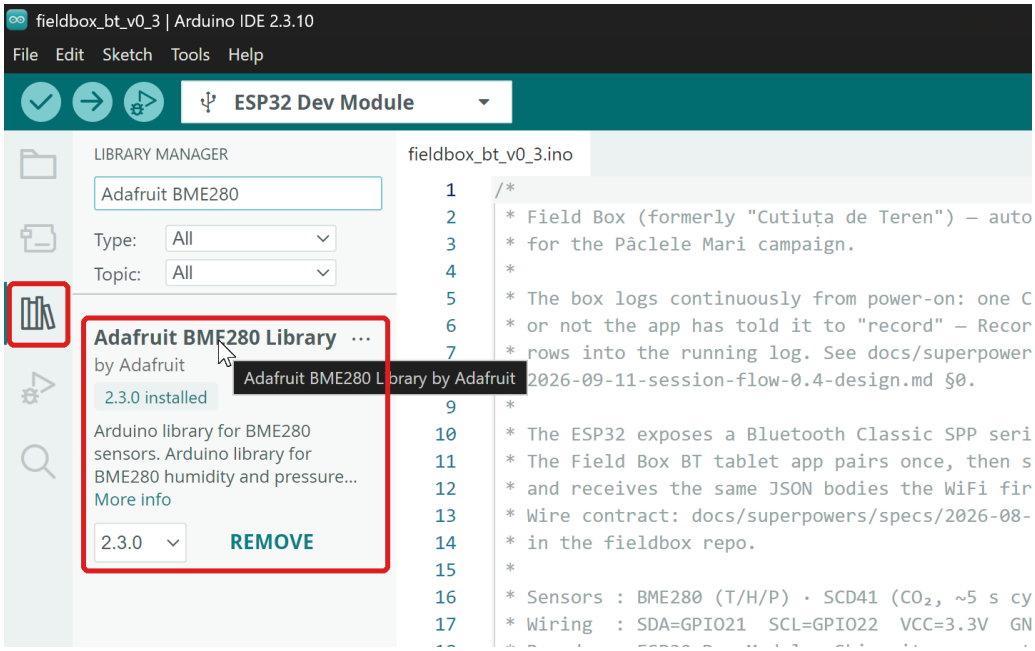


Boards Manager: esp32 by Espressif Systems

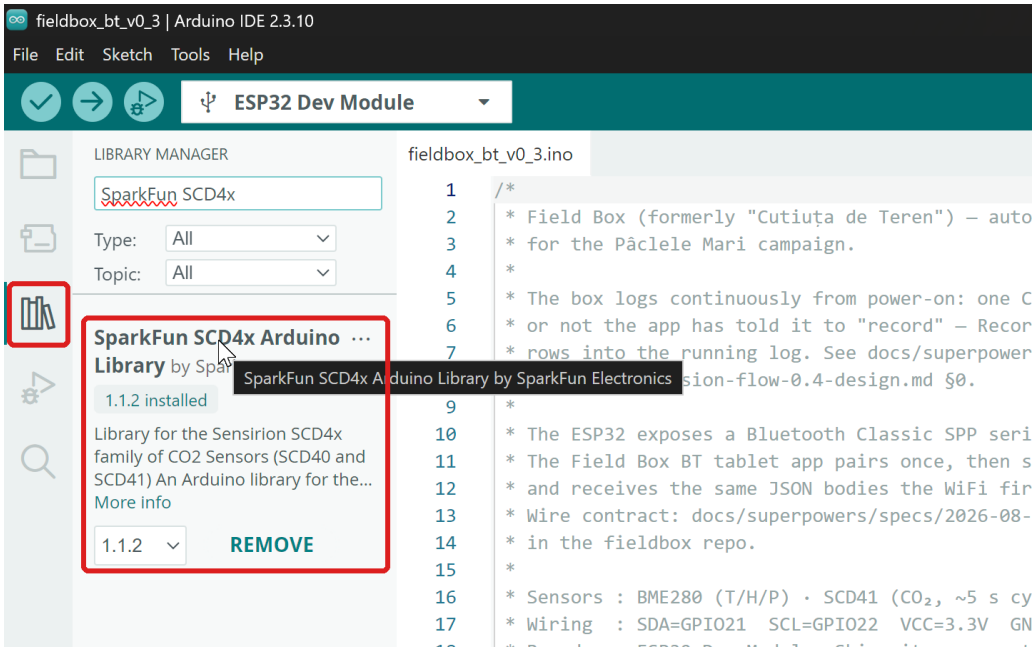
5. Click the third icon (**Library Manager**) and install these libraries one by one. Type the name and click **INSTALL**. If it asks about dependencies, click **INSTALL ALL**.

- **Adafruit BME280 Library**
- **Adafruit VEML7700 Library**
- **Adafruit Unified Sensor** (usually installed together with the first one)
- **SparkFun SCD4x Arduino Library**
- **BH1750** (by Christopher Laws)
- **Adafruit seesaw Library**
- **Adafruit SGP30 Sensor**
- **Sensirion Gas Index Algorithm**

The particle sensor's library is not in the Library Manager. Download the ZIP from **github.com/DFRobot/DFRobot_BMV080** (green **Code** button → **Download ZIP**), then in the IDE choose **Sketch → Include Library → Add .ZIP Library…** and pick the downloaded file. All these libraries are needed even if your box has none of the extra sensors.



Library Manager: Adafruit BME280

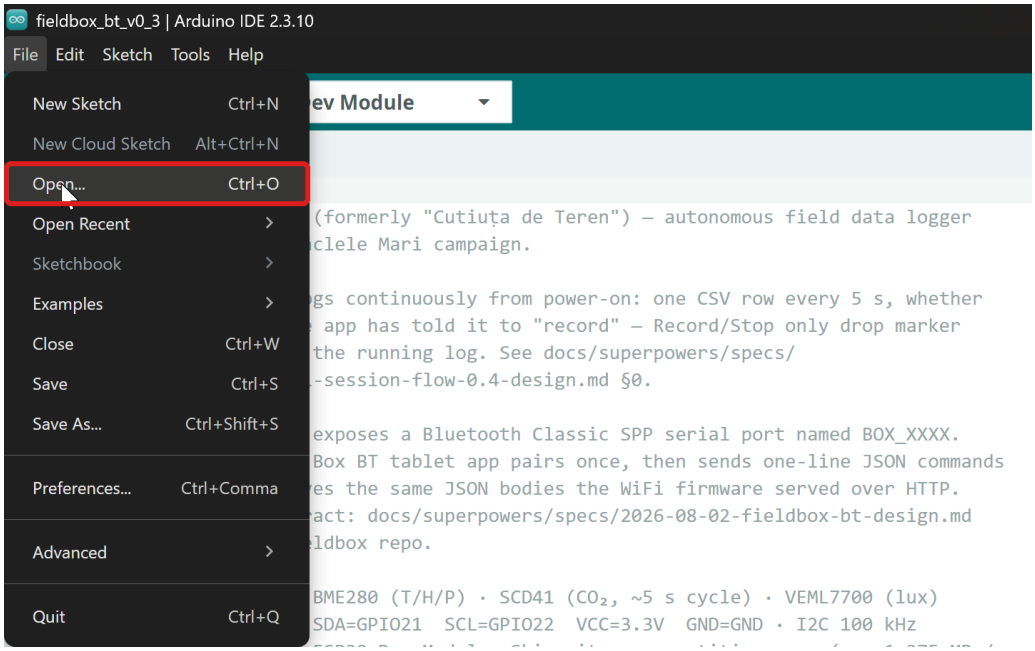


Library Manager: SparkFun SCD4x

5.3 Writing the firmware to the box

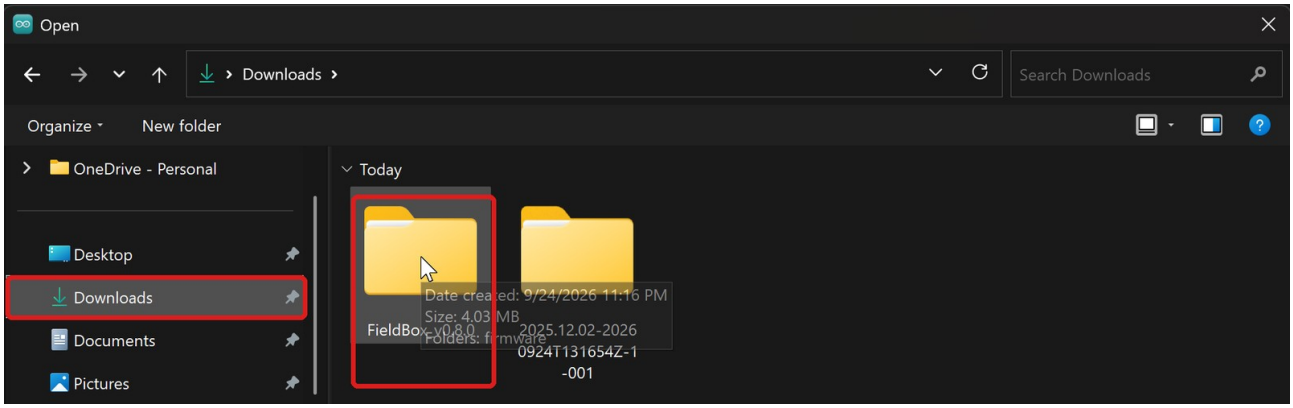
1. Plug the board inside the box into the computer with the USB cable.

2. In the Arduino IDE: **File → Open…**.



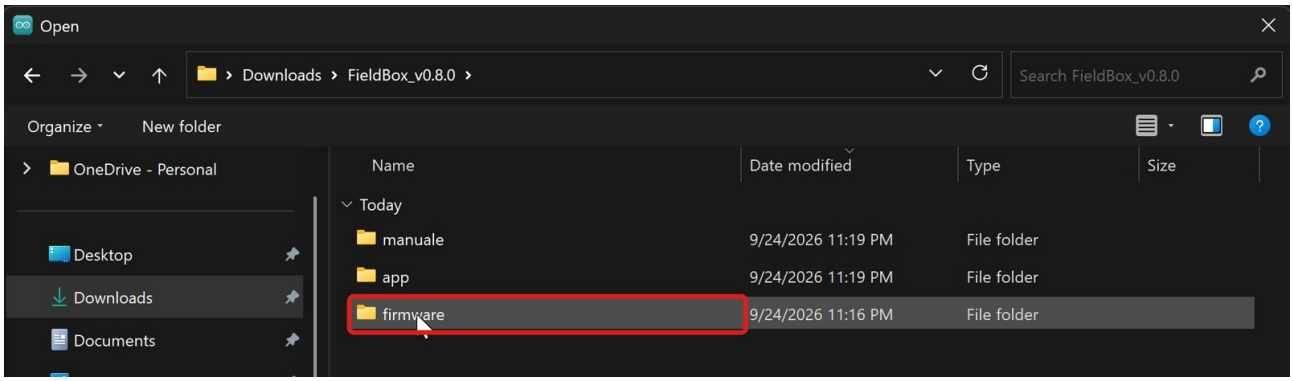
File → Open

3. In the **Open** window, pick **Downloads** on the left, then double-click the **FieldBox_v…** folder.



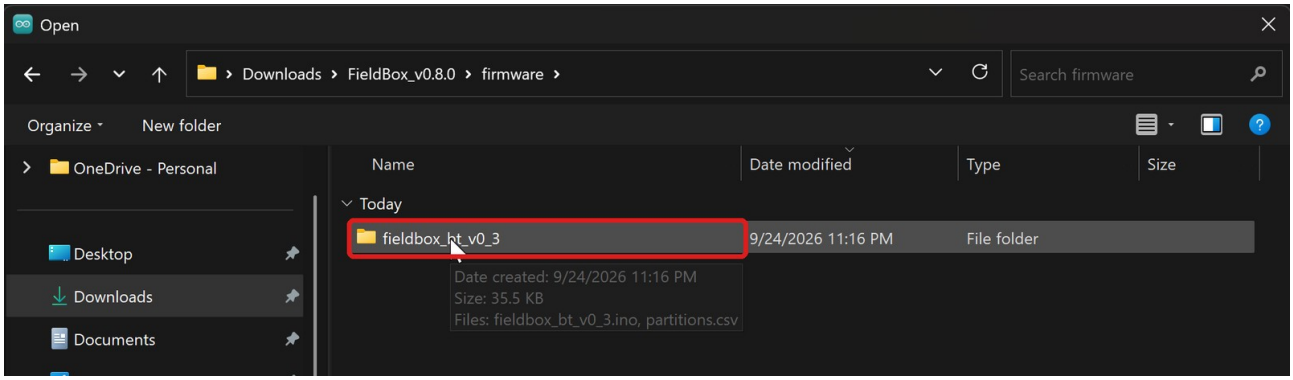
Downloads → FieldBox

4. Double-click **firmware**…



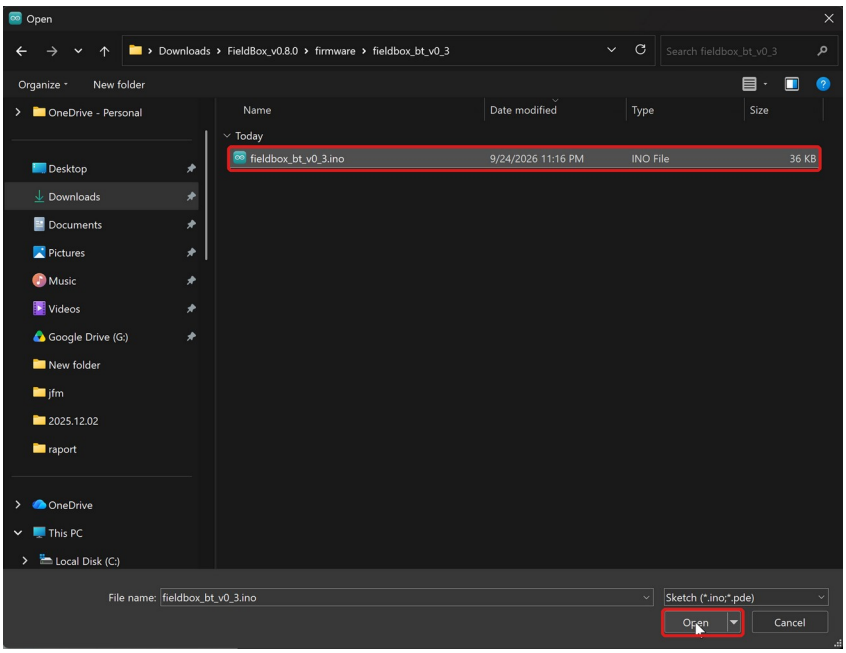
The firmware folder

5. ...then **fieldbox_bt_v0_4**.



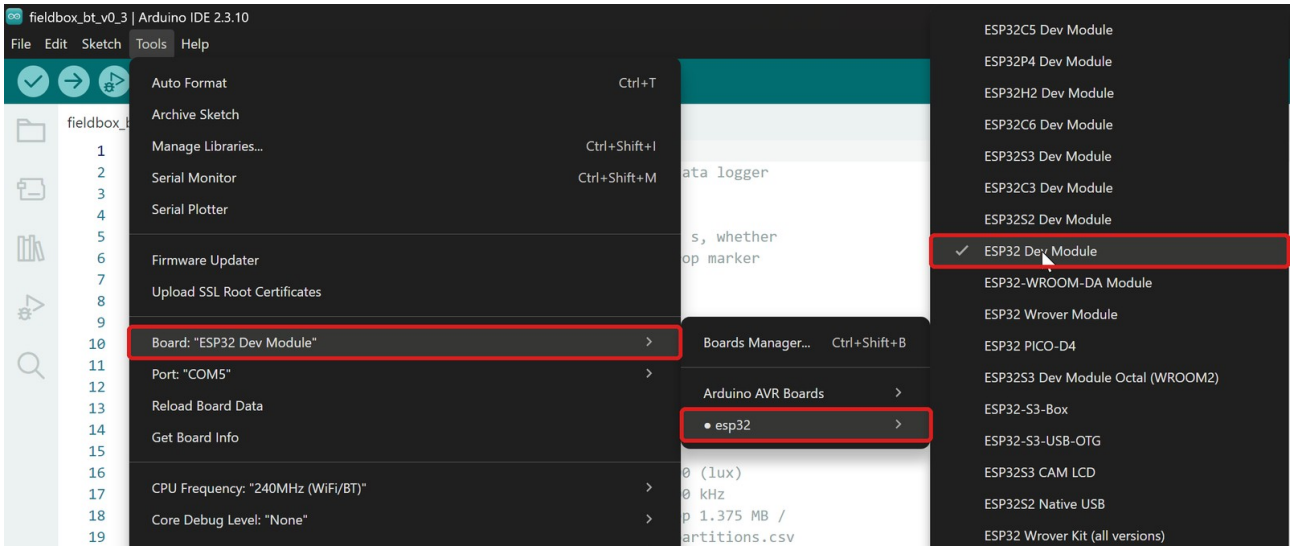
The fieldbox_bt_v0_4 folder

6. Select **fieldbox_bt_v0_4.ino** and click **Open**. The program opens in a new window. Carry on in that one, titled **fieldbox_bt_v0_4**.



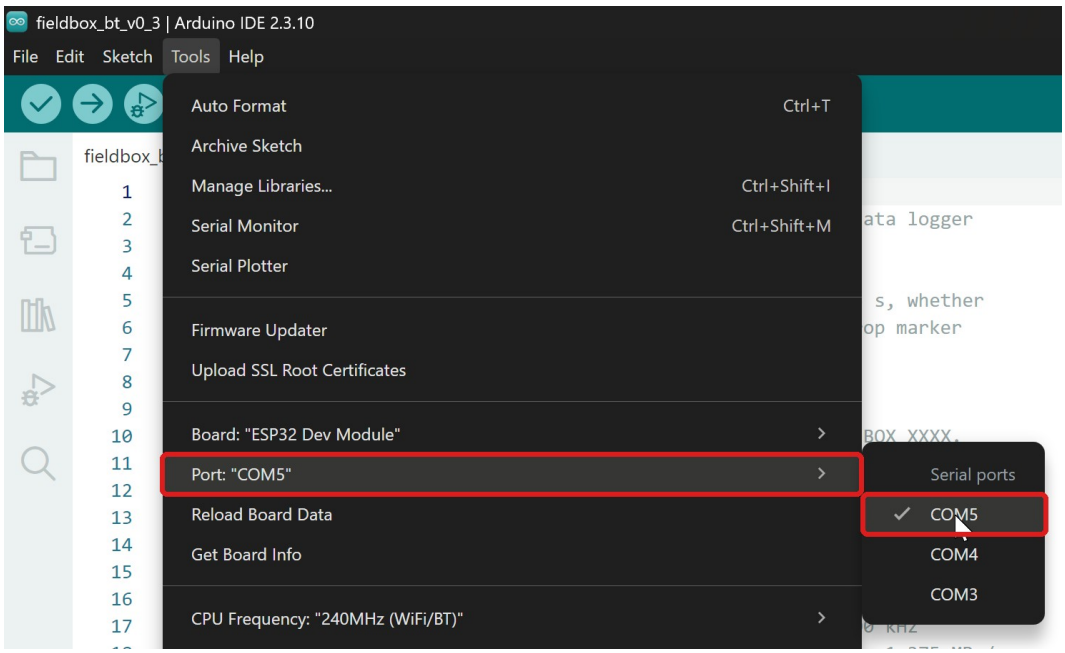
Opening fieldbox_bt_v0_4.ino

7. **Tools → Board → esp32 → ESP32 Dev Module**.



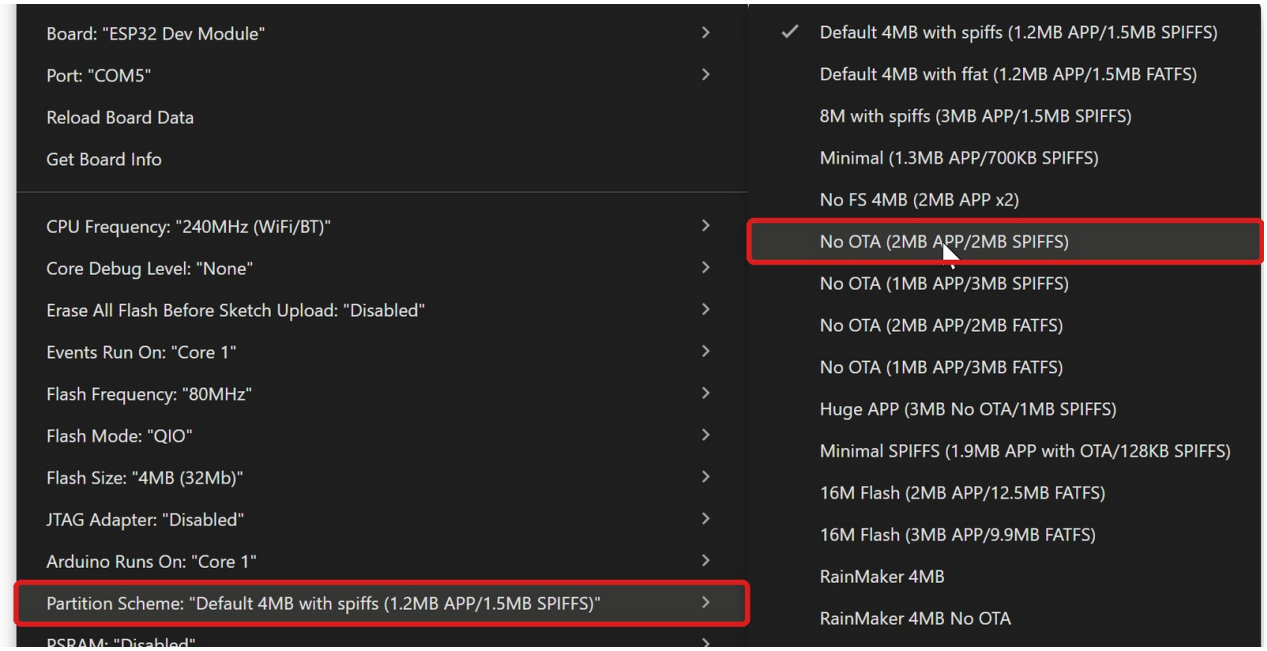
Tools → Board → esp32 → ESP32 Dev Module

8. **Tools → Port** → the board's port (e.g. **COM5**). If several ports are listed, unplug the cable and see which one disappears, then plug it back in. The port that comes back is the box.



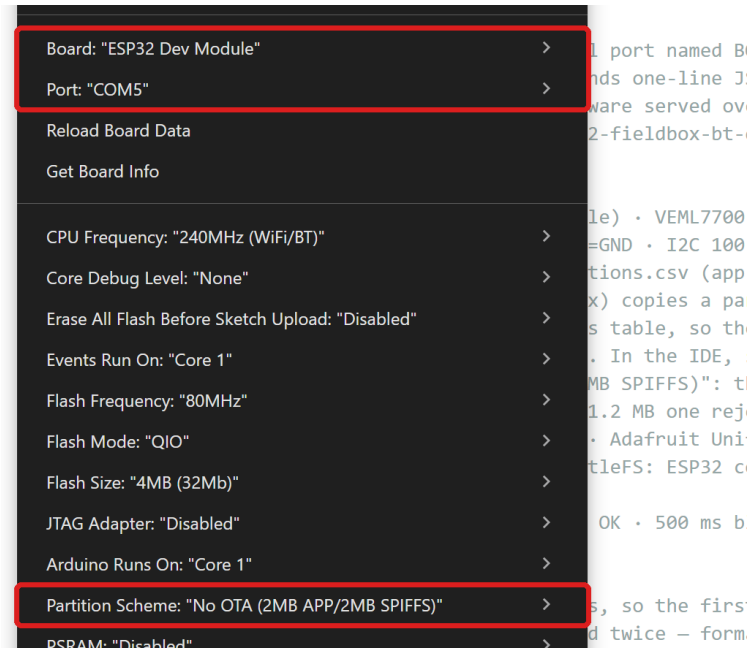
Tools → Port

9. **Tools → Partition Scheme → No OTA (2MB APP/2MB SPIFFS)**.



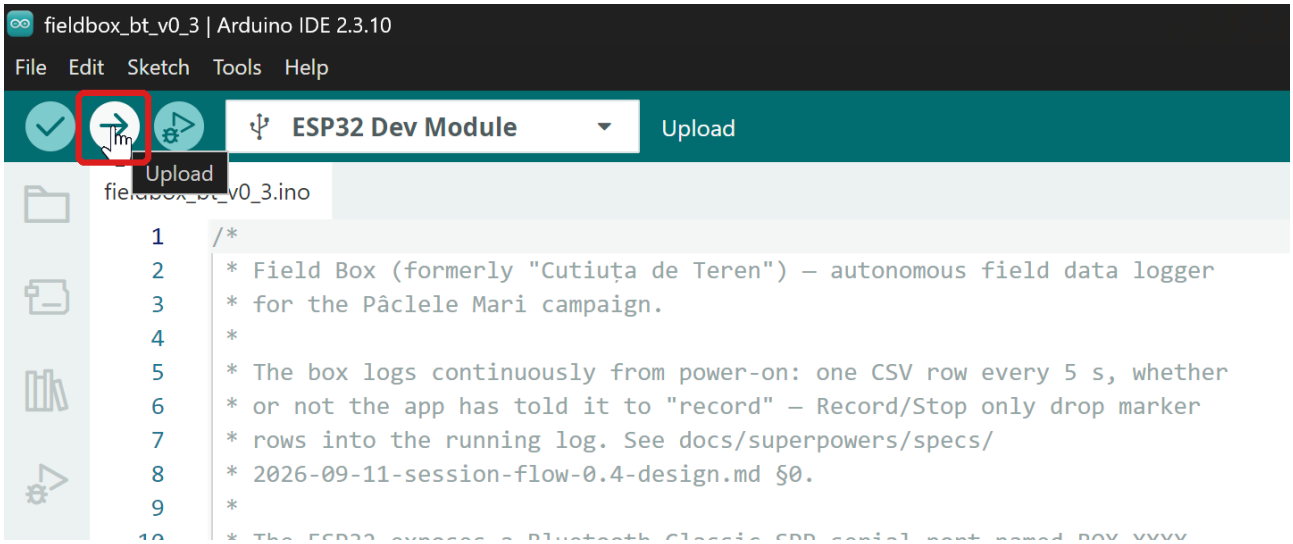
Tools → Partition Scheme → No OTA (2MB APP/2MB SPIFFS)

10. Open **Tools** once more and check the three lines: **Board: "ESP32 Dev Module"**, **Port: "COM…"** and **Partition Scheme: "No OTA (2MB APP/2MB SPIFFS)"**.



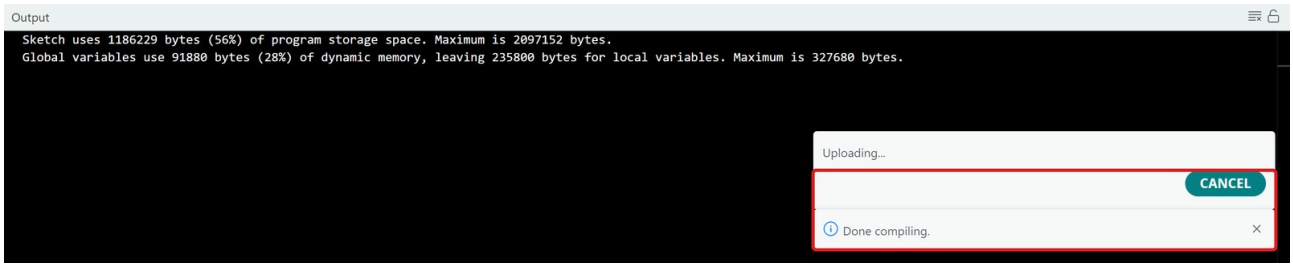
Checking the Tools settings

11. Click the round button with the right arrow (**Upload**).



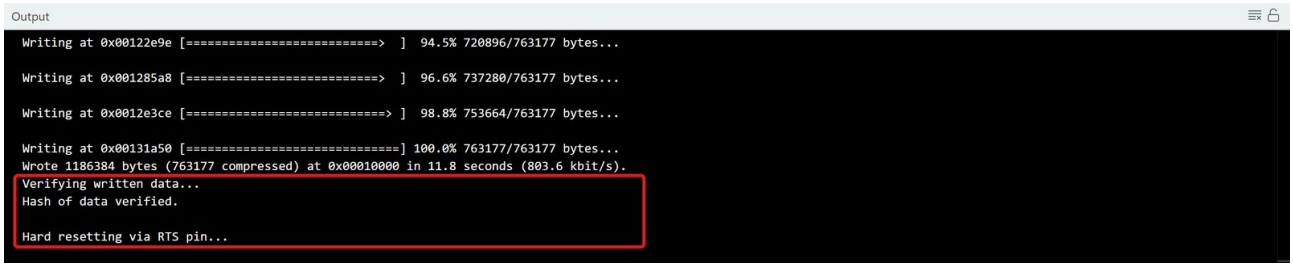
The Upload button

12. Wait. First **Compiling sketch…** (a few minutes the first time), then **Uploading…**, with **Writing at …** lines climbing to 100 %.



Compiling done, writing started

13. It is done when the bottom pane shows **Hash of data verified** and **Hard resetting via RTS pin…**. Unplug the USB cable.



Successful write

If the box had 0.3.x, its files stay; coming from 0.2, the first boot reformats the storage (a few seconds). Then the LED stays on for 2 seconds if the sensors are fine. In the app, the **Boxes** tab shows the new version (`firmware v0.4.1-bt`).

If it does not work:

- **No port under Tools → Port:** the cable is charge-only, or the CH340 driver is missing.

- **"Failed to connect… Wrong boot mode":** hold the **BOOT** button on the board while the bottom pane says **Connecting…**, then release it.
- **"Sketch too big": No OTA (2MB APP/2MB SPIFFS)** was not selected at step 9.
- **"…h: No such file or directory":** one of the libraries from step 5 of §5.2 is missing.

Other notes:

- The box's name (`**BOX_xxxx**`) comes from the board's Bluetooth MAC address and does not change with a firmware update.
- The package also has the ready-built `**firmware/fieldbox-fw-<version>.bin**`, for those who use **esptool** or the **Flash Download Tool**. It is written at address `**0x0**`.
- A box on **0.2.0-bt** still works with the 0.11.0 app, but without continuous background logging (the log only runs while a recording is active). Mixed fleets (some boxes updated, some not) are fine.

6. Uninstalling / data

- **Uninstalling the app** does not delete the files in **Documents/FieldBox** and **DCIM/FieldBox** — they stay on the phone.
- However, sessions imported for comparison, in **Documents/FieldBox/compare/**, are lost — on reinstalling, the pupils' ZIPs must be sent again.
- **To free up space:** from the **Files** tab, download and then delete old files from the box; on the phone, delete past days' files from **Documents/FieldBox** once they have been copied to a laptop.